

LANDER 11 SSE, WY, USA

WMO: 725730

Lat: 42.6754N

Lon: 108.6686W

Elev: 5773

StdP: 11.88

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94078

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.8	-0.2	-12.2	3.5	-4.4	-6.8	4.7	2.1	27.6	40.6	23.2	40.3	2.1	N/A	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 27.4	(c) 91.4	(d) 56.1	(e) 88.8	(f) 55.8	(g) 86.1	(h) 55.6	(i) 60.2	(j) 80.7	(k) 59.0	(l) 79.9	(m) 57.8	(n) 78.9	(o) 7.6	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.6	75.8	64.0	51.5	70.1	64.9	49.1	64.2	65.5	29.2	80.5	28.3	79.6	27.4	78.8	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.6	15.7	13.0	-14.2	95.7	4.3	2.0	-17.3	97.2	-19.8	98.3	-22.2	99.5	-25.3	101.0		
			DB													
			WB	-13.6	63.3	4.0	1.2	-16.5	64.2	-18.8	64.9	-21.0	65.6	-23.9	66.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	24.2	26.6	37.1	43.2	51.7	63.6	72.3	69.7	60.7	46.1	35.1	23.6
	DBStd	19.14	10.71	11.37	9.80	8.84	8.94	7.76	5.46	5.86	8.56	10.03	10.66	10.98
	HDD50	3662	799	654	409	237	87	5	0	0	16	186	452	817
	HDD65	7376	1264	1074	865	655	420	116	10	28	178	587	897	1282
	CDD50	2301	0	0	9	32	139	413	693	609	336	65	5	0
	CDD65	538	0	0	0	0	6	73	238	172	48	1	0	0
	CDH74	6706	0	0	0	7	115	1062	2892	1981	631	18	0	0
	CDH80	2521	0	0	0	0	11	375	1235	737	162	1	0	0
Wind	WSAvg	4.9	4.1	5.0	5.7	5.6	5.2	5.6	4.9	4.9	5.1	4.6	4.6	4.2
Precipitation	PrecAvg	12.9	0.7	0.6	1.0	1.7	2.2	1.3	0.8	0.7	1.0	1.2	0.8	0.8
	PrecMax	20.0	1.5	1.6	3.7	4.8	6.0	4.7	2.3	2.1	3.3	3.7	1.9	2.1
	PrecMin	6.7	0.2	0.2	0.0	0.3	0.1	0.2	0.0	0.2	0.1	0.3	0.1	0.3
	PrecStd	3.2	0.3	0.4	0.7	1.1	1.7	1.1	0.6	0.4	0.7	0.8	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	55.1	67.6	73.7	81.7	92.5	96.0	92.4	88.3	76.6	64.8	52.7
		MCWB	37.4	39.8	44.9	48.3	51.2	55.5	56.9	56.1	55.9	51.1	44.9	39.1
	2%	DB	46.3	50.4	61.9	68.2	77.6	88.4	92.9	89.7	84.3	70.7	59.6	47.9
		MCWB	35.7	37.9	42.9	46.2	51.5	55.1	56.4	55.4	53.5	48.3	42.2	36.4
	5%	DB	43.2	46.6	57.2	63.7	72.9	84.5	89.8	87.2	81.4	66.3	55.5	44.4
		MCWB	34.0	35.8	40.9	44.4	50.4	54.5	56.4	55.0	52.4	46.8	40.6	35.2
	10%	DB	40.0	43.0	52.9	59.6	68.2	80.4	87.0	84.6	77.4	62.4	51.5	40.3
		MCWB	32.0	33.5	39.0	42.9	48.8	54.3	56.6	54.9	51.9	45.3	38.5	32.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.0	41.5	46.1	50.0	56.1	61.7	62.4	60.6	59.8	52.7	45.2	39.4
		MCDB	47.0	52.3	65.3	71.7	72.0	80.0	81.7	79.2	81.1	71.0	62.3	48.7
	2%	WB	36.4	39.0	43.7	47.3	53.2	59.2	60.4	59.0	56.5	50.4	42.4	37.1
		MCDB	45.0	50.9	60.5	65.9	69.5	78.9	81.7	79.5	74.1	68.1	57.5	47.0
	5%	WB	34.5	36.7	41.7	45.3	51.3	57.4	59.5	57.6	54.9	48.5	40.7	34.9
		MCDB	43.1	47.1	56.9	61.8	67.7	77.7	80.8	79.1	74.4	64.6	54.0	43.9
	10%	WB	32.1	34.2	39.8	43.5	49.7	55.7	58.4	56.6	53.3	46.3	38.5	32.2
		MCDB	39.7	43.6	53.1	58.7	65.1	75.0	80.0	78.8	72.8	61.6	50.1	40.1
Mean Daily Temperature Range	5% DB	MDBR	18.5	18.7	20.4	21.7	21.7	26.3	27.4	26.8	25.1	21.9	20.3	18.4
		MCDBR	22.0	22.4	25.7	28.0	28.1	30.8	30.0	29.8	28.7	27.2	25.0	23.1
		MCWBR	14.1	13.6	12.8	13.6	12.5	12.5	10.8	11.0	11.1	12.6	12.8	14.6
	5% WB	MCDBR	21.8	22.2	24.8	27.0	24.3	27.8	27.2	26.8	24.2	24.6	24.9	23.1
		MCWBR	14.5	13.7	12.8	13.2	12.0	13.0	10.6	10.2	9.7	11.7	13.5	15.1
Clear-Sky Solar Irradiance	taub		0.234	0.240	0.260	0.305	0.317	0.313	0.340	0.331	0.298	0.265	0.244	0.235
	taud		2.519	2.524	2.519	2.411	2.440	2.478	2.428	2.444	2.532	2.601	2.576	2.523
	Ebn at Noon		299	313	317	306	303	303	293	294	298	299	293	288
	Edh at Noon		21	25	29	35	35	34	36	34	29	23	20	19
All-Sky Solar Radiation	RadAvg		685	1003	1407	1760	2027	2407	2318	2024	1644	1122	752	557
	RadStd		44	63	83	101	166	147	96	62	111	72	56	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page